

Software Requirements Specification: Advanced Todo List (Python)

1. Project Overview

A lightweight, desktop-based Todo List application designed for efficiency. The application features a persistent local database, flexible due date parsing, and a customizable GUI (Transparency, Mini-mode).

2. Technical Stack

- **Language:** Python 3.x
- **GUI Framework:** PyQt6 (Preferred for modern UI) or Tkinter with CustomTkinter .
- **Database:** SQLite3 (Standard library, no external installation required).
- **OS Support:** Windows (Primary focus for "Startup" feature).

3. Functional Requirements

3.1 Task Management

- **Add Task:**
 - Triggered by a "Add" button, opening a modal Dialog.
 - Fields: Title (Required), Description (Optional), Due Date (Optional), Is_Starred (Boolean).
- **Due Date Parsing:**
 - **Format A (MM/DD):** Automatically infer the year. If MM/DD is earlier than today, set to next year; otherwise, current year.
 - **Format B (YYYY/MM/DD):** Explicit date.
 - **Format C (Empty):** Persistent as NULL (Lowest priority).
- **Task Operations:**
 - **Complete:** Mark as done, record completion_time , and move to history.
 - **Delete:** Permanently remove from the database.
 - **Star/Unstar:** Toggle importance.

3.2 Sorting Logic (Strict Hierarchy)

Tasks in the main list must follow this order:

1. Is_Starred == True (Sorted by Due Date within this group).
2. Is_Starred == False (Sorted by Due Date ascending).
3. Tasks with no Due Date are always at the bottom.

3.3 Views & History

- **Main View:** Interactive list of active tasks with checkboxes.
- **Detailed/History View:**
 - **List Mode:** Displays all completed tasks with completion timestamps.
 - **Calendar Mode:** A monthly grid showing tasks based on their Due Date or Completion Date .

4. System Settings & UI Features

- **Transparency:** A slider to adjust the main window's opacity (0.1 to 1.0).
- **Always on Top:** Toggle `WindowStaysOnTopHint` .
- **Auto-Startup:** Option to add the script/executable to the Windows Registry (`HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run`).
- **Mini-mode:**
 - **Dimensions:** Fixed size (approx. 100px Width x 300px Height).
 - **Content:** Prioritize displaying the most urgent tasks (based on Sorting Logic).
 - **Interaction:** The list must be **Scrollable** to allow viewing lower-priority tasks within the restricted area.
 - **Behavior:** Maintain the `Topmost` (`WindowStaysOnTopHint`) attribute.

5. Database Schema (SQLite)

- **Table:** `tasks`
 - `id` : INTEGER PRIMARY KEY
 - `title` : TEXT
 - `due_date` : DATE (ISO format)
 - `is_starred` : INTEGER (0 or 1)
 - `status` : TEXT ('active', 'completed')
 - `created_at` : TIMESTAMP
 - `completed_at` : TIMESTAMP (NULL if active)

6. Non-Functional Requirements

- **Persistence:** Data must save automatically upon any change.
- **UX:** Single-click to star/unstar; double-click to view details.